



The Effects of Deadline Pressure on Attitudinal Ambivalence

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Abstract

This article examines how attitudinal ambivalence, derived from the positive and negative evaluations associated with an attitude toward a behavior, that has an explicit deadline, fluctuates as a deadline becomes imminent. Using a longitudinal design, it was shown that as a deadline for performing a behavior becomes imminent, the level of ambivalence associated with the behavior increased significantly. Such an increase was not realized for a control group that was assessed longitudinally but without the presence of an imminent deadline. In addition, we provide evidence that the increase in ambivalence as a deadline approaches is driven primarily by the increased salience of negative factors associated with the behavior.

Keywords: ambivalence, time horizon, blood donation

Attitudinal ambivalence is defined as the presence of both favorable and unfavorable components of an attitude (Scott, 1968), or the extent to which one's reactions to an attitude object are evaluatively mixed and characterized by both positive and negative elements (Breckler, 1994; Wegner et al., 1995). Underlying this conceptualization of attitudinal ambivalence is the notion that an individual can potentially hold both a positive and a negative evaluation of an attitude object simultaneously. Understanding attitudinal ambivalence is important to marketers because of the implication that a consumer who holds a neutrally-valenced attitude is not necessarily indifferent, and thus potentially responsive to positive marketer communications. Rather, the consumer may be ambivalent, consequently holding negative perceptions about the attitude object that must be addressed.

Conceptually, attitudinal ambivalence is typically characterized as a property of an attitude (e.g., Scott, 1968). Attitudinal ambivalence is believed to be negatively associated with other attitudinal properties such as affective-cognitive consistency, attitude extremity (Thompson et al., 1995) and certainty (Gross et al., 1995). In addition, a highly ambivalent attitude has been purported to roughly obtain the same consequences of a weak versus a strong attitude suggesting a negative correlation between ambivalence and attitude strength (Thompson et al., 1995; Armitage and Conner, 2000).

Research on attitudinal ambivalence has focused primarily on two aspects: (1) its assessment (e.g., Kaplan, 1972; Breckler, 1994; Thompson et al., 1995); and (2) its relationship with regard to attitudes and other attitudinal properties (e.g., Jonas et al., 1997; Armitage and Conner, 2000; Sparks et al., 2001). What has not been explored directly are the dynamic processes that may cause levels of attitudinal ambivalence to fluctuate over

time. This article is concerned with the issue of how ambivalence may fluctuate as the positive and negative evaluations associated with the attitude toward a behavior become relatively more or less salient. This issue is important to marketers because attitudinal ambivalence has been shown to affect the attitude-behavioral intention relationship (Moore, 1973; Sparks et al., 2001).

1. The Independence of Positive and Negative Attitudinal Components

An attitude has typically been conceptualized and measured as the evaluation of some object or behavior along a single bipolar dimension ranging from maximally negative to maximally positive (see Eagly and Chaiken, 1993). Under such a conceptualization, as the evaluation toward an object or behavior increases in positivity (negativity), it is assumed conceptually, and required operationally to automatically, and to the exact same degree, decrease in negativity (positivity); that is, positive and negative elements are reciprocally activated (Kaplan, 1972; Cacioppo et al., 1997).

Some researchers (e.g., Scott, 1968; Kaplan, 1972; Thompson et al., 1995), however, have theorized that the positive and negative evaluations on which an attitude is based, although typically negatively correlated, are actually conceptually independent. Intuitive support for the independence of positive and negative evaluations is not difficult to invoke; for example, a person may view their boss as both tough (negative) but fair (positive); a consumer may view a product as both high in quality (good) but over-priced (bad); a person may view donating blood as a worthy behavior (good) but may be afraid of needles (bad); etc. The independence of positive and negative evaluations has been demonstrated empirically as well. For example a number of researchers have found correlations between positive and negative attitudinal components to be in the -0.13 to 0.40 range (Kaplan, 1972; Thompson et al., 1995; Jonas et al., 1997).

Eagly and Chaiken (1995) provide a theoretical framework generally supportive of treating positive and negative evaluative components of an attitude as independent. They propose that both the overall evaluation of an attitude object, and the components on which the overall evaluation is based, are linked together in memory as part of an overall intra-attitudinal cognitive structure associated with the attitude object or behavior. This is consistent with Pratkanis and Greenwald's (1989) model of attitude structure in long-term memory in which an evaluative summary of an object or behavior is supported by a knowledge structure that can include both judgments in favor of, or against, the overall evaluative summary (see also Judd and Brauer, 1995 for a similar conceptualization of intra-attitudinal cognitive structure). Thus, although positive and negative evaluations underlying a particular attitude associated with an attitude object or behavior may often be highly and negatively correlated (i.e., reciprocally activated), there is no compelling theoretical or empirical rationale to require them, by definition, to be characterized this way. This theoretical framework provides the foundation for the exploration of attitudinal ambivalence.

2. Imminent Deadlines and Ambivalence

Many behaviors are associated with an explicit deadline. For example, registering to vote and voting are both behaviors that are associated with deadline dates; gifts and cards must be purchased prior to birthdays and anniversaries; charities such as the United Way typically have cutoffs associated with their fundraising campaigns; blood drives typically have a specified period in which the blood mobile will be available, etc.

Research suggests that attitudes toward volitional behaviors associated with an explicit deadline may be different than attitudes toward behaviors without a deadline. For example, Wright and Weitz (1977) hypothesize and provide evidence that individuals become more loss averse as a loss possibility becomes more temporally imminent. They reason that individuals do not like to think about negative aspects of a decision, and thus tend to discount them when the decision is temporally distant. Consistent with this rationale is the finding that individuals are averse to negative outcomes in the distant future because of the corresponding feelings of dread associated with anticipating the negative outcome over time (Loewenstein and Prelec, 1991). Thus, it seems reasonable that individuals would avoid thinking about negative aspects in order to avoid feelings of dread over time.

Cacioppo and his colleagues (Cacioppo et al., 1997; Cacioppo and Bernston, 1999), using as a foundation the literature on conflict in general, and approach-avoidance behavior more specifically, posit and provide empirical evidence of a “positivity offset” and a “negativity bias” associated with the positive and negative evaluations of an attitude object. A positivity offset suggests that when a goal (e.g., performing a behavior) is distal, positive aspects of the behavior will be more salient and thus predominate negative aspects of the goal. A negativity bias suggests that a single negative aspect relating to a behavior is given more weight, with regard to an overall evaluation, than a single positive aspect. Thus, as a goal approaches, negative aspects associated with the behavior are given marginally more weight than the existing positive aspects, rendering the negative aspects associated with the behavior increasingly salient.

As it relates to behaviors associated with a deadline and ambivalence, the extant research suggests that when a deadline associated with a behavior is relatively far off, although both positive and negative elements associated with a behavior may exist (e.g., knowledge and experience), positive aspects related to the behavior would be more salient, resulting in relatively low levels of ambivalence toward the behavior. Conversely, as the goal becomes more proximate, negative aspects associated with the behavior would become increasingly salient, resulting in relatively higher levels of ambivalence associated with the behavior. This reasoning is consistent with Sparks et al.’s (2001) proposition that ambivalence may change over time as a result of fluctuations in the salience of positive and negative components associated with a behavior (i.e., “temporal instability” p. 57). Further support of the effect of increased salience of negative aspects associated with a behavior comes from Priester and Petty (1996) who posit that conflicting (i.e., negative) aspects associated with a behavior are the primary drivers of attitudinal ambivalence. Thus, if negative components of a behavior, such as they exist, are given increasing weight as a deadline associated with a behavior approaches, and given that positive and negative aspects of the behavior may

be held simultaneously, the level of attitudinal ambivalence associated with a volitional behavior should increase as a deadline becomes imminent.

Thus:

H1: As the deadline associated with a volitional behavior containing both positive and negative evaluative components becomes imminent, attitudinal ambivalence associated with the behavior will increase over time.

H2: The increase in ambivalence, as the deadline associated with a volitional behavior containing both positive and negative evaluative components becomes imminent, will be driven by an increase in the salience of the negative elements associated with the behavior over time.

Ambivalence has been conceptualized as an attribute of an attitude that moderates the relationship between attitudes and behavioral intent. For example, Moore (1973) found that ambivalence moderated the attitude–behavioral intention relationship such that those with high ambivalent scores indicated a significantly lower attitude–behavioral intention correlation than those with low ambivalent scores. More recently, both Armitage and Conner (2000) and Sparks et al. (2001) also found that less ambivalent attitudes were more predictive of behavioral intentions than highly ambivalent attitudes. If attitudes higher in ambivalence have been shown to be less predictive of behavioral intentions than attitudes lower in ambivalence, in light of H1, it follows that the differential levels of ambivalence over time, driven by the presence or absence of an imminent deadline, and associated with a volitional behavior, will moderate the relationship between attitudes toward the behavior and behavioral intentions as the deadline becomes imminent:

H3: As the deadline associated with a volitional behavior, containing both positive and negative evaluative components, becomes imminent, attitudes will become less predictive of behavioral intentions.

3. Method

To test the hypothesis that ambivalence would increase as a deadline becomes imminent required the selection of a behavior that would be likely to include both positive and negative elements. It was believed that the behavior of donating blood would satisfy this criterion and thus it was selected as the target behavior.

3.1. Design, Subjects, and Procedure

Longitudinal data were collected at two time periods (waves) from two independent samples. Thus, we utilized a 2 (distant deadline/imminent deadline) $\times$ 2 (wave 1/wave 2)

mixed-factor design. The deadline factor was manipulated between subjects and the wave factor was measured within subjects. All subjects were college students who participated in the study in exchange for extra credit.

During the last week in September of 2000, subjects in the distant-deadline condition were told that one of the colleges at the university was sponsoring a blood drive on campus in March of 2001, approximately 26 weeks away. During these sessions, subjects were administered a questionnaire that would assess their overall attitudes, positive and negative evaluations, and behavioral intentions relating to the March 2001 blood drive. Upon completion of the questionnaire, subjects signed up for the second wave and were then dismissed.

Sessions for wave 2 of the distant-deadline condition were held six weeks later. Once again, subjects were told of a university-sponsored blood drive that was being held in March of 2001, now approximately 20 weeks away. During wave 2, subjects in the distant-deadline condition completed the same questionnaire that they completed in the first wave. Sixty-eight subjects participated in both waves of the distant-deadline condition.

During the second week in February, the study was run again with a new, entirely independent sample of subjects (the imminent-deadline condition). The same procedures used for wave 1 in the distant-deadline condition were used in wave 1 of the imminent-deadline condition; that is, subjects were once again told that the university was sponsoring a blood drive in March. Subjects in the imminent-deadline condition were asked to complete the same questionnaire that was used during wave 1 within the distant-deadline condition. As with the distant-deadline condition, subjects in the imminent-deadline condition were asked to again complete the same questionnaire six weeks later. However, in wave 2 of the imminent-deadline condition, subjects were told that the blood drive was being held on the Friday of the week the wave-2 questionnaires were being administered. Sixty-seven subjects participated in both waves of the imminent-deadline condition.

3.2. *Assessing and Computing Ambivalence*

Conceptualizing attitudinal ambivalence as the extent to which both positive and negative evaluative components of the overall attitude are relatively strong and relatively equal requires: (1) the independent assessment of the positive and negative elements associated with an attitude object along separate bivalent scales; and (2) the combining, using some computational formula, of the positive and negative elements into a single ambivalence measure (Kaplan, 1972; Breckler, 1994; Thompson et al., 1995).

The procedure used for independently assessing the positive and negative evaluative components of an attitude was first proposed by Kaplan (1972), and has been used by many researchers since (Breckler, 1994; Thompson et al., 1995; Priester and Petty, 1996; Jonas et al., 1997; Bromer, 1998; Armitage and Conner, 2000; Sparks et al., 2001). Specifically, positive aspects toward donating blood were assessed using a three-item scale in which subjects were told to ignore the unfavorable characteristics, feelings of dissatisfaction, and the harmful qualities related to donating blood, and to only consider the favorable qualities, feelings of satisfaction, and beneficial qualities of donating blood. The three-item scale was

anchored by 1 = not at all favorable, 9 = very favorable; 1 = not at all satisfied, 9 = very satisfied; and 1 = not at all beneficial, 9 = very beneficial. The reliability of this scale was $\alpha_{\text{wave 1}} = 0.79$; $\alpha_{\text{wave 2}} = 0.79$. These items were averaged to form a single measure of positive aspects concerning donating blood.

Negative aspects toward donating blood were assessed using a three-item scale in which subjects were told to ignore the favorable characteristics, feelings of satisfaction, and the beneficial qualities related to donating blood, and only to consider the unfavorable qualities, feelings of dissatisfaction and harmful qualities of donating blood. The three-item scale was anchored by 1 = not at all unfavorable, 9 = very unfavorable; 1 = not at all dissatisfied, 9 = very dissatisfied; and 1 = not at all harmful, 9 = very harmful. The reliability of this scale was $\alpha_{\text{wave 1}} = 0.82$; $\alpha_{\text{wave 2}} = 0.78$. These items were averaged to form a single measure of negative aspects concerning donating blood. Positive and negative aspects concerning donating blood were separated by a number of questions on the questionnaire, as recommended by Thompson et al. (1995).

To combine the positive and negative evaluative components into a single ambivalence measure, we use the computation suggested by Thompson et al. (1995); this computation has been used often in subsequent research (e.g., Armitage and Conner, 2000; Sparks et al., 2001). For ease of interpretation, three was added to the equation to make the measure range from 0 (minimum ambivalence) to 12 (maximum ambivalence):

$$\text{Ambivalence} = \{[(\text{Positive} + \text{Negative})/2] - |\text{Positive} - \text{Negative}|\} + 3. \quad (1)$$

3.3. Other Measures

3.3.1. Attitude toward donating blood The positive and negative evaluations associated with an object or behavior are purported to underlie the expression of an attitude regarding the particular object or behavior (Cacioppo et al., 1997). Thus, within the intra-attitudinal cognitive structure associated with an object or behavior, the positive evaluations, negative evaluations, and overall attitude constructs are conceptualized as uniquely represented in memory. Thus, in addition to measuring the positive and negative components that determine the overall attitude, a measure of overall attitude was obtained as well (cf. Sparks et al., 2001; Armitage and Conner, 2000; Bromer, 1998; Jonas et al., 1997).

Attitudes toward donating blood at the blood drive in March were assessed using a four-item, nine-point semantic differential scale anchored in response to the behavior being 1 = bad, 9 = good; 1 = harmful, 9 = beneficial; 1 = undesirable, 9 = desirable; 1 = worthless, 9 = valuable. The reliability of this scale was $\alpha_{\text{wave 1}} = 0.72$; $\alpha_{\text{wave 2}} = 0.76$. These items were averaged to form a single measure of attitude toward the behavior of donating blood. This attitude measure was used to calculate attitude extremity (discussed later in the article); operationalized as the absolute value of a study participant's deviation from the midpoint (i.e., 5) on this overall attitude measure (cf. Krosnick et al., 1993).

3.3.2. Behavioral intentions concerning donating blood Behavioral intentions were measured on a two-item scale. The first item was a six-point scale anchored by +3 = definitely will/ -3 = definitely will not donate blood at the blood drive. The second item

was the statement: "I intend to donate blood at the University's blood drive in March," and was anchored by 1 = strongly disagree and 9 = strongly agree. Reliability of this scale was $r_{\text{wave 1}} = 0.90$, $r_{\text{wave 2}} = 0.89$.

4. Results

The focus of the analyses is on repeated-measures analysis of variance. In these analyses, the distant-deadline condition will act as a control group for comparison to the imminent-deadline condition; recall that the time lag between wave 1 and wave 2 is the same for both the distant- and imminent-deadline conditions.

4.1. Attitudinal Ambivalence

H1 predicts that as the deadline associated with a volitional behavior becomes imminent, attitudinal ambivalence will increase, within subject, over time. This hypothesis implies that if the deadline is not imminent (i.e., is still in the relatively distant future) ambivalence will not increase, or at least not as much as when the deadline is imminent. To test this hypothesis, a 2×2 mixed factor ANOVA was conducted with time (i.e., wave 1 and wave 2) as a within-subject, repeated-measures factor, condition (distant- versus imminent deadline) as a between-subjects factor, and attitudinal ambivalence as the dependent variable. This analysis indicated a significant effect due to the within-subjects, time factor ($F(1, 133) = 9.17$, $p < 0.01$, $\eta^2 = 0.065$), but a nonsignificant interaction effect on time due to the deadline manipulation for attitudinal ambivalence ($F(1, 133) = 0.918$, $p > 0.30$, $\eta^2 = 0.007$). However, due to the directional nature of our hypotheses, a priori predictions were further tested (Kirk, 1982). Thus, we conducted separate repeated-measures ANOVAs among subjects in both the distant- and imminent-deadline conditions using our measure of ambivalence at wave 1 and wave 2 as the repeated-measures variable. These analyses indicated that although the level of ambivalence within the distant-deadline condition did increase, this increase was not statistically significant from wave 1 ($M_{\text{wave 1}} = 4.32$) to wave 2 ($M_{\text{wave 2}} = 4.89$; $F(1, 67) = 2.21$, $p > 0.10$) and that the strength of association of wave 1 to wave 2, in the distant-deadline condition, was rather low ($\eta^2 = 0.032$). However, as predicted, within the imminent-deadline condition, the level of ambivalence did increase significantly from wave 1 ($M_{\text{wave 1}} = 5.05$) to wave 2 ($M_{\text{wave 2}} = 6.15$; $F(1, 66) = 7.71$, $p < 0.01$), and the strength of association was greater compared to the distant-deadline condition ($\eta^2 = 0.105$), in support of H1 (see Table 1).

The above findings are supported by nonparametric statistics that show that two thirds (67.2%) of participants in the imminent-deadline condition experienced an increase in ambivalence. This percentage was significantly higher than the 51.5% of participants that experienced an increase in ambivalence in the distant-deadline condition ($z = 1.94$, $p = 0.05$).

These results suggest that as a deadline for engaging in a behavior, characterized by some level of ambivalence, approaches, the level of ambivalence associated with that behavior increases. Further, as the deadline becomes imminent, this effect becomes more

Table 1. Study Results

	Wave 1	Wave 2
Distant-deadline condition		
Attitudinal ambivalence	4.32	4.89
Positive aspects of behavior	8.00	7.65 ^a
Negative aspects of behavior	3.75	3.97
Attitude–intention correlation	0.23	0.28
Imminent-deadline condition		
Attitudinal ambivalence	5.05	6.15 ^{b,c}
Positive aspects of behavior	7.85	7.51 ^a
Negative aspects of behavior	4.16	4.83 ^{a,c}
Attitude–intention correlation	0.52 ^{a,c}	0.31

^a Significant (within-subject) difference between wave 1 and wave 2 ($p < 0.05$).

^b Significant (within-subject) difference between wave 1 and wave 2 ($p < 0.01$).

^c Significant (between-subject) difference between distant-deadline condition and imminent-deadline condition ($p \leq 0.05$).

pronounced as evidenced by the fact that the effects of the passage of time on ambivalence in the imminent-deadline condition are stronger compared to the distant-deadline condition.

4.2. Positive and Negative Aspects of Ambivalence

H2 predicts that the increase in ambivalence, as a deadline associated with a volitional behavior becomes imminent, will be driven by an increase in the negative aspects associated with the behavior. Again, the implication is that if a deadline is not imminent, there will be no shift in the positive or negative aspects of the behavior from wave 1 to wave 2. To test this hypothesis we again conducted a 2×2 mixed factor ANOVA with time as a within-subject, repeated-measures factor, condition as a between-subjects factor, and positive aspects associated with the behavior as the dependent variable. This analysis again indicated a significant effect due to the within-subjects, time factor ($F(1, 133) = 8.71$, $p < 0.01$, $\eta^2 = 0.061$), but a nonsignificant interaction effect on time due to the deadline manipulation for positive aspects of the behavior ($F(1, 133) = 0.007$, $p > 0.90$, $\eta^2 = 0.000$). To test our specific, a priori hypotheses, separate repeated measures ANOVAs were again conducted among subjects in both the distant- and the imminent-deadline conditions, using positive aspects of the behavior as the dependent variable. The results were somewhat surprising in that positive aspects associated with the behavior demonstrated a significant decline in both the distant-deadline condition ($M_{\text{wave 1}} = 8.00$; $M_{\text{wave 2}} = 7.65$; $F(1, 67) = 4.24$, $p < 0.05$) and the imminent-deadline condition ($M_{\text{wave 1}} = 7.85$; $M_{\text{wave 2}} = 7.51$; $F(1, 66) = 4.51$, $p < 0.05$), with a very similar strength of association in both conditions ($\eta^2 = 0.060$, $\eta^2 = 0.064$ in the distant- and imminent-deadline conditions, respectively). However, although positive aspects associated with the behavior did decrease over time in each condition, there was no relative effect of positive elements

within the imminent deadline condition (i.e., no effect on the target condition relative to the control condition).

Next, a 2×2 mixed factor ANOVA was conducted with time as a within-subject, repeated-measures factor, condition as a between-subjects factor, and negative aspects associated with the behavior as the dependent variable. This analysis again indicated a significant effect due to the within-subjects, time factor ($F(1, 133) = 6.64, p = 0.01, \eta^2 = 0.048$), but a nonsignificant interaction effect on time due to the deadline manipulation for negative aspects of the behavior ($F(1, 133) = 1.64, p > 0.20, \eta^2 = 0.012$). To test our specific, a priori hypotheses, separate repeated measures ANOVAs were again conducted among subjects in both the distant- and the imminent-deadline conditions, using negative aspects of the behavior as the dependent variable. The analysis indicated that the negative aspects associated with the behavior within the distant-deadline condition did not increase significantly from wave 1 ($M_{\text{wave 1}} = 3.75$) to wave 2 ($M_{\text{wave 2}} = 3.97; F(1, 67) < 1$), and that the strength of association of wave 1 to wave 2, in the distant-deadline condition, was low ($\eta^2 = 0.013$). However, within the imminent-deadline condition, the negative aspects of the behavior did increase significantly from wave 1 ($M_{\text{wave 1}} = 4.16$) to wave 2 ($M_{\text{wave 2}} = 4.83; F(1, 66) = 6.89, p = 0.01$), and the strength of association was greater compared to the distant-deadline condition ($\eta^2 = 0.095$).

By definition, a change in the level of ambivalence associated with engaging in a behavior can be driven by a change in either the positive or negative aspects associated with the behavior, or both. The question remains, however, as to which contributes more to the variability in ambivalence over time. The relationship between attitudinal ambivalence and the positive and negative aspects associated with behavior was assessed by conducting multiple regression with the change in ambivalence from wave 1 to wave 2 as the dependent variable and with the change in the positive and negative aspects associated with the behavior as the independent variables. The overall model was significant ($F(2, 132) = 427.94, p = 0.000$). Additionally, both the change in positive and the change in the negative aspects associated with the behavior were significant in predicting the change in attitudinal ambivalence. Importantly, the standardized beta coefficient for the change in negative aspects associated with the behavior (standardized $b_{\text{negatives}} = 0.939$) was more than six times greater than the standardized beta coefficient for the change in positive aspects associated with the behavior (standardized $b_{\text{positives}} = 0.152$).

These results suggest that the significant decline in ambivalence, as the deadline for engaging in the behavior approaches, is driven by both a decrease in the positive aspects associated with the behavior and an increase in the negative aspects associated with the behavior. Further, the finding that positive aspects associated with the behavior decline about equally across the two conditions, combined with the finding that negative aspects associated with the behavior increase significantly only in the imminent-deadline condition and not the distant-deadline condition, suggest that the stronger effect relating to the increase in ambivalence in the imminent- compared to the distant-deadline condition is driven primarily by changes in the negative aspects associated with the behavior, in support of H2.

4.3. *Attitude–Behavioral Intentions Relationship*

Previous research has demonstrated that the relationship between attitudes and behavioral intent is weaker (stronger) for attitudes associated with high (low) levels of ambivalence. The logic of H3 is that if ambivalence increases over time, as was evidenced in the imminent-deadline condition, the attitude–intention relationship, within subject, should be stronger for wave 1 when ambivalence is relatively lower, than for wave 2 when ambivalence is relatively higher. Further, if ambivalence does not increase over time, as was evidenced in the distant-deadline condition, the strength of the attitude–intention relationship should be the same in both waves. More specifically, it is hypothesized that in the distant-deadline condition, the attitude–intention correlation will not change from wave 1 to wave 2. However, it is hypothesized that in the imminent-deadline condition, the attitude intention correlation will be greater in wave 1 than in wave 2. To test the differences across waves, procedures for testing differences in correlation coefficients measured within-subject were used (Steiger, 1980). For the distant-deadline condition, the difference in the attitude–intention correlation between wave 1 (0.23) and wave 2 (0.28) was not significant ($z < 1$). However, within the imminent-deadline condition, the attitude–intention correlation was significantly greater in wave 1 (0.52) than in wave 2 (0.31; $z = 1.81$; $p < 0.05$). Thus, at wave 2, the relationship between attitude and behavioral intent weakens relative to wave 1 in the imminent-deadline condition, but not in the distant-deadline condition, as predicted by H3.

4.4. *Exploration of Effects on Attitudinal Extremity*

Original ambivalence research was driven by a desire to better understand and interpret the meaning of a mid-point response on a bipolar attitudinal scale (Scott, 1968; Kaplan, 1972). Thus ambivalence and attitude extremity (defined as the extent to which an individual's attitude deviates from the midpoint of the favorableness/unfavorableness scale (Krosnick et al., 1993)) have typically been conceptualized as negatively correlated (Thompson et al., 1995). However, what has not been explored are the effects of attitude extremity, over time, given the presence of an imminent deadline.

Based on the results associated with ambivalence reported thus far, one might predict that attitude extremity would remain unchanged over time in the distant-deadline condition, in association with the relatively low amount of change in ambivalence in that condition. And in fact, there was no change, over time, in attitude extremity in the distant-deadline condition ($F(1, 67) = 1.41$, $p > 0.25$, $\eta^2 = 0.021$). Further, one might predict a decrease in attitude extremity, over time, in the imminent-deadline condition in association with the increase in ambivalence in that condition. However, there was no change, over time, in attitudinal extremity in the imminent-deadline condition ($F(1, 66) < 1$, $\eta^2 = 0.001$).

This finding has a two interesting implications. First, the findings suggest that individuals can experience attitudinal ambivalence at various points along the attitudinal scale, not just at the midpoint, a finding consistent with Thompson et al.'s (1995) speculation about the range of attitudinal ambivalence. Second, these findings suggest that the imminent

deadline can affect qualitative dimensions of attitudes (e.g., ambivalence, attitude strength, etc.) while leaving quantitative dimensions (e.g., valence, extremity, etc.) unaffected. Further research should be done to explore these two implications.

5. General Discussion

5.1. *Summary and Interpretation of the Findings*

The purpose of this study was to examine the dynamic effects of an explicit deadline on individuals' attitudinal ambivalence toward a behavior, and to implicate the processes by which individual-level changes in attitudinal ambivalence occur. This study expands the boundary on ambivalence research by proposing and testing the idea that an individual's level of attitudinal ambivalence toward a behavior will fluctuate if factors exist to modify the salience of the positive and negative components of an attitude. In this study, we proposed that the arrival of an imminent deadline would be such a factor.

We found that as a deadline for engaging in a behavior became imminent, individuals experienced an increase in the level of attitudinal ambivalence associated with the behavior. By including a control group in our longitudinal design, we were able to rule out the alternative explanations that ambivalence increased either as simply a function of the passage of time, or that our experimental procedures alone produced the change in ambivalence over time. Additionally, our findings suggest that the small increase in ambivalence observed in the distant-deadline condition was driven primarily by a decrease in the positive aspects associated with the behavior, and that the significant increase in ambivalence in the imminent-deadline condition was driven by both a decrease in the positive aspects and an increase in the negative aspects associated with the behavior. Finally, analyses to test H3 suggest a weakening of the relationship between attitudes toward the behavior and the intention to perform the behavior as an explicit deadline becomes imminent.

These findings have implications that are applicable in more typical marketing contexts as well. For example, sales promotion activities, such as sales, coupons, or give-aways, are often associated with explicit deadlines indicating when the promotional activity will end. It is likely that the consideration of purchase within a product category, especially, because of the cost and potential risk, a durable good, could produce ambivalence as the deadline for the expiration of the sales promotion becomes imminent. The findings suggest that as an expiration period approaches, a marketer would want to shift focus from delineating the positive aspects of the product to actively counter-arguing any negatives associated with the purchase of the product. For example, if cost was a primary consideration for the purchase of a particular durable good, even with the terms of the limited-time sale, a marketer could reframe the issue, as the expiration date for the sales promotion became more imminent, and focus discussion of the cost in terms of the cost per day rather than the total cost (cf. Gourville, 1998).

5.2. *Limitations and Future Research*

Our findings demonstrate how the approach of an explicit deadline can affect the relative salience of the positive and negative factors associated with a behavior. However, the approach of an imminent deadline may not be the only factor that will affect the relative relationship between positive and negative factors; other factors that can affect the relative salience of positive or negative aspects of a behavior should also impact the level of attitudinal ambivalence toward a specific behavior. Future research should explore additional cues that may trigger the relative accessibility of these positive and negative aspects of a behavior and their effects on attitudinal ambivalence.

In both the distant- and imminent-deadline conditions, the date for the blood drive was given as March. This was done to enhance the credibility of the cover story because the university was actually sponsoring a blood drive in March. However, there is a potential confound with the time of data collection in that there may be differences in collecting data in September 2000 versus collecting data in March 2001. Future research should control for the time of data collection.

Finally, in the current research, ambivalence was measured and not manipulated. Future research should focus on ways to manipulate ambivalence to ensure these reported relationships hold up under more controlled conditions. Specifically, future research should attempt to tease apart the effect of ambivalence and deadline imminence on the attitude-behavioral intention relationship.

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